

Quadratic chirps are not trace-class Fourier–Wigner multipliers

Candidate partial result for arXiv:2502.16248

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Status. Candidate substantial partial result, likely valid, requiring human review. The source’s explicit nonmeasure candidate is ruled out quantitatively. The full classification of trace-class Fourier–Wigner multipliers remains open.

1 Source question and result

For $z = (x, \xi) \in \mathbb{R}^{2d}$, let $\rho(z)$ be the symmetric time–frequency shift and let

$$\mathcal{F}_W(T)(z) = \text{tr}(T\rho(-z))$$

be the Fourier–Wigner transform. For $m \in L^\infty(\mathbb{R}^{2d})$, the associated Fourier–Wigner multiplier is formally

$$\mathfrak{T}_m(T) = \mathcal{F}_W^{-1}(m\mathcal{F}_W(T)).$$

The algebra $\mathfrak{M}_1(\mathbb{R}^{2d})$ consists of those m for which this map is bounded on the trace class $\mathcal{S}^1$.

Samuelsen proves that $\mathcal{F}_\sigma(\mathcal{M}(\mathbb{R}^{2d})) \subseteq \mathfrak{M}_1$ and asks whether equality holds [1, Question 1]. The paper singles out

$$s(z) = \sin(\pi|z|^2)$$

as a possible counterexample: $\mathcal{F}_\sigma(s)$ is not a finite measure, but the source leaves membership of s in $\mathfrak{M}_1$ undecided. The theorem below rules out this candidate and, more generally, every nonconstant radial quadratic chirp.

REFERENCES

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Fourier transform of complex Gaussians. The derivatives of $\sin(\pi|z|^2)$ are of at most polynomial growth, and thus, for any Schwartz function $\Phi \in \mathcal{S}(\mathbb{R}^{2d})$, it follows that

$$\left| z^\alpha \frac{\partial^\beta}{\partial z^\beta} \Phi(z) \sin(\pi|z|^2) \right| \leq \sum_{j \leq |\beta|} |z^\alpha P_j(z) \Phi^{(j)}(z)|,$$

for any $\alpha, \beta \in \mathbb{N}_0^{2d}$, where P_j is a polynomial for each j . This shows that $\Phi(z) \sin(\pi|z|^2)$ is a Schwartz function whenever $\Phi \in \mathcal{S}(\mathbb{R}^{2d})$. From here we conclude that

$$\Phi * \tau = \mathcal{F}_\sigma(\mathcal{F}_\sigma(\Phi) \sin(\pi|\cdot|^2)) \in \mathcal{S}(\mathbb{R}^{2d}) \subseteq L^1(\mathbb{R}^{2d}),$$

but $\tau \notin \mathcal{M}(\mathbb{R}^{2d})$. However, it is unclear whether or not $\sin(\pi|z|^2)$ is a trace class multiplier. This question is equivalent to asking if mixed-state localisation operator with symbol τ given by (10) is a trace class operator for any choice of window functions $f, g \in L^2(\mathbb{R}^d)$ by Lemma 5.1. We therefore end with an open question.

Question 1. For each $m \in \mathfrak{M}_1$, does there exist $\mu \in \mathcal{M}(\mathbb{R}^{2d})$ such that $m = \mathcal{F}_\sigma(\mu)$?

Figure 1. The proposed sine candidate and Question 1 in [1, p. 21], cropped from the local source PDF.

2 Idea of the proof

Test the multiplier on rank-one projections generated by increasingly squeezed Gaussians. Their Fourier–Wigner transforms are anisotropic Gaussians of fixed L^2 mass. Multiplication by a unimodular quadratic chirp therefore leaves the Hilbert–Schmidt norm of the output equal to one.

On the other hand, the output integral kernel can be evaluated exactly. It is narrow in the difference variable $t - s$, with width of order $r^{-1/2}$. Schur’s test makes its operator norm decay like $r^{-d/2}$. Since

$$\|A\|_2^2 \leq \|A\|_\infty \|A\|_1,$$

fixed Hilbert–Schmidt mass and vanishing operator norm force the trace norm to grow like $r^{d/2}$. For the sine, the two conjugate chirps cannot cancel: an exact oscillatory Gaussian integral leaves a uniformly positive amount of Hilbert–Schmidt mass.

3 The squeezed-Gaussian calculation

For $r > 0$, define the normalized Gaussian

$$g_r(t) = (2r)^{d/4} e^{-\pi r |t|^2}, \quad t \in \mathbb{R}^d,$$

and the rank-one operator $G_r = g_r \otimes g_r$. Thus $\|G_r\|_1 = 1$. Direct calculation with the source conventions gives

$$\mathcal{F}_W(G_r)(x, \xi) = \exp \left[-\frac{\pi}{2} \left(r|x|^2 + r^{-1}|\xi|^2 \right) \right]. \quad (1)$$

We use the following centered-kernel formula. If $F = \mathcal{F}_W(A)$ and K_A is the integral kernel of A , then

$$K_A(t, s) = \int_{\mathbb{R}^d} F(t - s, \xi) e^{2\pi i \xi \cdot (t+s)/2} d\xi. \quad (2)$$

This follows immediately by inserting the definition of $\rho(x, \xi)$; it is also the usual ambiguity-function/kernel relation.

Lemma 1 (Chirp kernel bound). *Let $c \in \mathbb{R} \setminus \{0\}$ and $q_c(z) = e^{\pi i c |z|^2}$. Put $A_{r,c} = \mathfrak{T}_{q_c}(G_r)$, initially interpreted in $\mathcal{S}^2$. Then*

$$\|A_{r,c}\|_2 = 1, \quad (3)$$

$$\|A_{r,c}\|_\infty \leq \left(\frac{2}{|c|r} \right)^{d/2}. \quad (4)$$

Proof. Equation (3) follows from the unitarity of $\mathcal{F}_W : \mathcal{S}^2 \rightarrow L^2(\mathbb{R}^{2d})$, the unimodularity of q_c , and $\|\mathcal{F}_W(G_r)\|_2 = \|G_r\|_2 = 1$.

For the operator norm, set

$$D_{r,c} = \frac{1}{2r} - ic, \quad u = \frac{t+s}{2}, \quad x = t-s.$$

Applying (2) to $q_c \mathcal{F}_W(G_r)$ and evaluating the Gaussian Fourier integral gives, with the square-root branch determined by $\operatorname{Re} D_{r,c} > 0$,

$$K_{r,c}(t, s) = D_{r,c}^{-d/2} \exp \left(-\frac{\pi |u|^2}{D_{r,c}} - \frac{\pi r}{2} |x|^2 + \pi i c |x|^2 \right). \quad (5)$$

Since $\operatorname{Re}(D_{r,c}^{-1}) > 0$ and $|D_{r,c}| \geq |c|$,

$$|K_{r,c}(t, s)| \leq |c|^{-d/2} e^{-\pi r|t-s|^2/2}.$$

Therefore

$$\sup_t \int_{\mathbb{R}^d} |K_{r,c}(t, s)| ds, \quad \sup_s \int_{\mathbb{R}^d} |K_{r,c}(t, s)| dt \leq |c|^{-d/2} \left(\frac{2}{r}\right)^{d/2}.$$

Schur's test proves (4). □

Theorem 2 (Quadratic chirp obstruction). *For every $c \in \mathbb{R} \setminus \{0\}$,*

$$q_c(z) = e^{\pi i c |z|^2} \notin \mathfrak{M}_1(\mathbb{R}^{2d}).$$

More precisely, whenever $A_{r,c}$ is trace class,

$$\|A_{r,c}\|_1 \geq \left(\frac{|c|r}{2}\right)^{d/2}. \quad (6)$$

Proof. For any trace-class operator A , its singular values give $\|A\|_2^2 \leq \|A\|_\infty \|A\|_1$. Lemma 1 therefore yields (6). Since $\|G_r\|_1 = 1$ and the right-hand side tends to infinity, $\mathfrak{T}_{q_c}$ cannot be bounded on $\mathcal{S}^1$. □

4 The source's sine candidate

Theorem 3 (Negative decision for the proposed candidate). *Let $s(z) = \sin(\pi|z|^2)$. Then*

$$s \notin \mathfrak{M}_1(\mathbb{R}^{2d}).$$

Quantitatively, if $B_r = \mathfrak{T}_s(G_r)$ and B_r is trace class, then

$$\|B_r\|_1 \geq \frac{1 - 5^{-d/2}}{2} \left(\frac{r}{2}\right)^{d/2}. \quad (7)$$

Proof. Because $s = (q_1 - q_{-1})/(2i)$, Lemma 1 gives

$$\|B_r\|_\infty \leq (2/r)^{d/2}. \quad (8)$$

By Plancherel and (1),

$$\|B_r\|_2^2 = \int_{\mathbb{R}^{2d}} \sin^2(\pi(|x|^2 + |\xi|^2)) e^{-\pi(r|x|^2 + r^{-1}|\xi|^2)} dx d\xi.$$

Using $\sin^2 \theta = (1 - \operatorname{Re} e^{2i\theta})/2$ and the complex Gaussian integral, this equals

$$\frac{1}{2} \left[1 - \operatorname{Re}((r - 2i)^{-d/2} (r^{-1} - 2i)^{-d/2}) \right]. \quad (9)$$

The oscillatory factor in braces has modulus

$$((r^2 + 4)(r^{-2} + 4))^{-d/4} = (17 + 4(r^2 + r^{-2}))^{-d/4} \leq 5^{-d/2}.$$

Hence $\|B_r\|_2^2 \geq (1 - 5^{-d/2})/2$. Combining this with (8) and $\|B_r\|_2^2 \leq \|B_r\|_\infty \|B_r\|_1$ proves (7), which diverges with r . □

Corollary 4. *No nonconstant finite linear combination of distinct radial quadratic chirps $z \mapsto e^{\pi i c_j |z|^2}$ belongs to $\mathfrak{M}_1(\mathbb{R}^{2d})$.*

Proof. Subtract the constant term, which is already a measure multiplier. For the remaining distinct nonzero c_j , the triangle inequality in (4) makes the output operator norm $O(r^{-d/2})$. Expanding the squared Hilbert–Schmidt norm produces diagonal terms $\sum_j |a_j|^2$; each cross term is a complex Gaussian integral with nonzero chirp parameter and tends to zero as $r \rightarrow \infty$. Thus the Hilbert–Schmidt norm stays bounded below, and the same singular-value argument forces trace-norm divergence. $\square$

5 Scope, upgrade attempts, and search status

Theorem 3 settles the concrete candidate immediately preceding the source’s Question 1, but it does not answer the universal classification. The strongest surviving reformulation is operator-algebraic: a completely bounded Weyl-covariant multiplier has a Fourier–Stieltjes symbol, so a full affirmative answer would follow if every bounded multiplier on $\mathcal{S}^1$ were automatically completely bounded. Covariance alone does not make this formal; finite two-torsion Weyl systems have diagonal transposition maps with bounded norm one and growing completely bounded norm.

Two further upgrade routes were tested. Compactly supported localization reduces to the classical L^1 multiplier theorem by the source’s main result, but its comparison constants grow under exhaustion. A Lagrangian-restriction approach likewise runs into the lack of uniform equicontinuity of the Fourier–Wigner images of the trace-class unit ball. The squeezed-Gaussian calculation identifies the same concentration obstruction quantitatively. The accompanying attempt record gives the detailed route audit.

Bounded novelty check. The run indexes, exact question, exact sine candidate, quadratic-chirp terms, Weyl-covariant-map terminology, twisted Fourier multipliers, and citing works were searched through 11 August 2026. No later full answer and no prior statement of Theorems 2–3 was found. The 1984 framework of Werner [2] characterizes positive covariant correspondence rules, not arbitrary bounded multipliers. This is a bounded novelty audit, not a definitive priority claim.

Human-review recommendation. First check the centered kernel convention in (2). Then audit the factor $D_{r,c}^{-d/2}$ and the two uses of $\|A\|_2^2 \leq \|A\|_\infty \|A\|_1$. Finally verify the modulus estimate after (9). These are the only normalization-sensitive points.

References

- [1] H. J. Samuelsen, *Fourier–Wigner multipliers and the Bochner–Riesz conjecture for Schatten class operators*, arXiv:2502.16248v2 (2025).
- [2] R. F. Werner, *Quantum harmonic analysis on phase space*, J. Math. Phys. **25** (1984), 1404–1411.